

REAL ESTATE PORTFOLIO INSIGHTS

Week 1 Internship Project — House Price Prediction | Analyst: Aastha Anshu | Date: June 19, 2026

Executive Summary

This report presents an analytical model to predict residential house prices using property attributes from a real-world housing dataset of 545 properties. By leveraging data preprocessing, feature engineering, and linear/non-linear machine learning algorithms, we successfully modeled pricing dynamics and isolated the highest-leverage value drivers for housing assets.

Model Performance & Comparison

Evaluation Metric	Linear Regression (Best Fit)	Random Forest Regressor
Mean Absolute Error (MAE)	\$970,043.40	\$1,022,560.05
Root Mean Squared Error (RMSE)	\$1,324,506.96	\$1,401,496.84
R ² Score (Variance Explained)	0.6529	0.6114

Key Insights & Value Drivers

- 1. Dominant Value Driver (Size):** Lot size (area) is the primary predictor of house prices. Every additional square foot significantly raises the property's baseline market value.
- 2. Functional Layout (Bathrooms vs. Bedrooms):** Surprisingly, the number of bathrooms has a much higher impact on price (14.6% feature importance) compared to bedrooms (6.4%). Homes with multiple stories also command a strong pricing premium.
- 3. High-Leverage Enhancements:** Modern installations, specifically airconditioning, show a powerful correlation with higher selling prices, reflecting buyer preference for turn-key comfort.
- 4. Model Selection Insight:** The Linear Regression model outperformed the Random Forest Regressor (R^2 of 0.65 vs 0.61). This indicates the housing price relationships are largely linear, and a simple parametric model generalizes better on this sample size without overfitting.

Business & Strategy Recommendations

Actionable Strategy for Real Estate Firms: Advise clients and developers to focus capital budgets on targeted improvements rather than expansion. Instead of adding a bedroom, converting underutilized space into an extra bathroom or retrofitting central air conditioning represents the highest return on investment (ROI). Furthermore, valuations should be anchored primarily on area and bathroom count rather than bedroom count.

Actual vs. Predicted House Prices

